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Experimental and thermodynamic assessment of the Ce–Zr system

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ABSTRACT

The phase equilibria of the binary Ce–Zr system have been investigated between 900 K and 1800 K by a combining conical nozzle levitation technique with *in situ* high energy synchrotron X-ray diffraction. From the diffraction data the phase sequences and the liquidus line were determined experimentally. The Ce–Zr system is an eutectic type. The eutectic reaction $L \leftrightarrow bcc\text{-Ce} + hcp\text{-Zr}$ occurs at $T = 1113 \pm 5$ K at the eutectic composition of 1.5 ± 1 at% Zr. Based on the experimental data a self-consistent thermodynamic description of Ce–Zr phase diagram is derived.

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1. Introduction

Rare earth-transition metals (RE-TM) are used as permanent magnets which derive their exceptional magnetic properties from the favourable combination of rare earth metals with transition metals. Partial replacement of Nd by Ce or Ce-based magnets are of current interest because natural abundance of Ce is about four times that of Nd [1]. Mixed oxides in the Ce–Zr–O system are important as an excellent promoter for the development of automotive catalysts [2,3]. For further improvement of the properties by designing the microstructure the knowledge of thermodynamic data is indispensable. Calculation of the corresponding phase diagrams by the CALPHAD method [4] is a powerful tool for the optimization of alloys and thermal processing. The thermodynamic description of such multi-component systems requires the knowledge of the parameters of all subsystems. For the binary Ce–Zr system a hypothetic monotectic phase diagram is reported in [5] which is also quoted in [6,7] together with an evaluation of the published experimental data. The work of Harris and Raynor [8] is considered as the only reliable information concerning the Ce–Zr system data by Palenzona and Cirafici [7]. Experimental data on the solid solubility of Ce in *hcp*-Zr are given as determined from lattice parameters of quenched samples, and a very low concentration of Zr in Ce is concluded [8]. No reliable experimental information exist so far for the phase equilibria above the melting temperatures and the reported phase diagram is at least questionary. A new determination of the Ce–Zr system was therefore already suggested in [6] considering the limited and rather old works available.

Here we report on a new experimental assessment of the Ce–Zr system. We will show that the presentation given in [5–7] is incorrect. By probing the phase equilibria at high temperatures we observe clear evidence of the eutectic character of the Ce–Zr system.

2. Experimental

Master alloys of $Ce_{100-x}Zr_x$ ($x = 5, 10, 15, 60, 85, 90, 95$) with a mass of about 20 g were prepared from pure elements 99.98 wt% Zr (HWN titan GmbH), and 99.9 wt% Ce (Goodfellow) in an arc furnace, on a water-cooled copper hearth under purified Ti-gettered argon atmosphere. The buttons were remelted in a Hukin-type cold-crucible induction melting facility and cast into a copper mould with a 6 mm

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inner diameter. Thermal analysis was carried out by differential scanning calorimetry (DSC, Netzsch) with a heating rate of 20 K/min. The microstructure was analysed by scanning electron microscopy (SEM) in the backscattered electron (BSE) mode using a Gemini–Zeiss microscope equipped with an energy-dispersive spectrometer (Bruker AXS) for composition analysis. The chemical composition of the phases was determined at 15 keV using the intensities of the Ce L- and Zr L-emission lines.

For *in situ* X-ray diffraction investigations, alloy ingots of about 2 mm in diameter (40 mg) with nominal compositions $Ce_{40}Zr_{60}$ were prepared by arc-melting of proper quantities of pure Ce and Zr (Alfa Aesar 99.99 wt% purity) under an argon atmosphere with a Ti getter. The ingots were inverted and remelted several times to ensure compositional homogeneity. The difference in sample mass after arc-melting was less than 0.1%. For calibration also pure elements Ce and Zr samples were prepared as 2 mm spheres-like samples.

In situ high energy synchrotron X-ray diffraction (XRD) was performed in transmission geometry (wave length $\lambda = 0.01093$ nm) at the beam-line BL04B2 of SPring-8 in Japan. The samples were processed by a conical nozzle levitation using filter-purified Ar-6.0 gas. A CO_2-laser has been applied for heating the samples (Synrad, firestar f200). More details on the used set-up can be found in [9]. A recent review reports about the different types of levitation techniques and their uses for studying structure and thermophysical properties of liquids [10]. During the experiments, the levitated sample was repeatedly heated above its liquidus temperature ($T_{liquidus}$) and subsequently cooled down passing solidification and solid-state transformations to ensure reproducibility of the measurements. All samples had metallic shiny surfaces and no remarkable change of the weight after the levitation experiments. The temperature was measured by a two-colour pyrometer. The pyrometer values were corrected by adapting the temperature of the first XRD pattern with *bcc*-Zr and liquid phases ($T_{XRD} = 1263$ K) to the corresponding DSC event ($T_{DSC} = 1113$ K) according to $T_{corr} = 1/((1/T_{pyro}) + (1/T_{DSC}) - (1/T_{XRD}))$. The scattered intensity was continuously recorded with acquisition time of 0.6 s by a two-dimensional X-Ray Image Intensifier with a coupled CCD-camera during heating or cooling. The scattering intensity $I(q)$ as a function of the wave vector q ($q = 4\pi \sin\theta/\lambda$) was obtained by azimuthally integration of the original 2D images using a Fit2D software package [11].

Fig. 1 shows a typical temperature–time profile of a measurement with two heating and cooling segments of the $Ce_{40}Zr_{60}$ alloy.

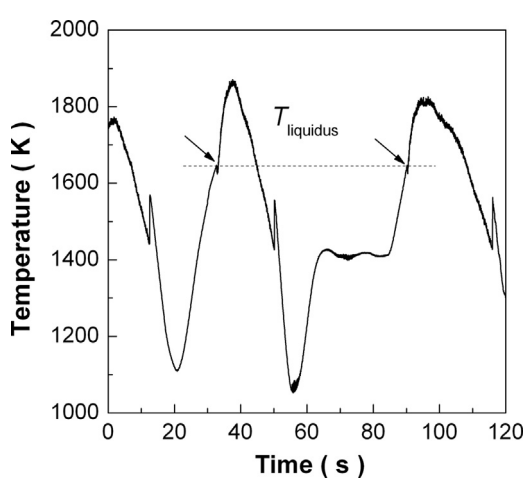

Fig. 1. Temperature profile of a $Ce_{40}Zr_{60}$ sample during levitation. Simultaneously XRD patterns were recorded with an acquisition time of 0.6 s. Two heating and cooling cycles are shown.

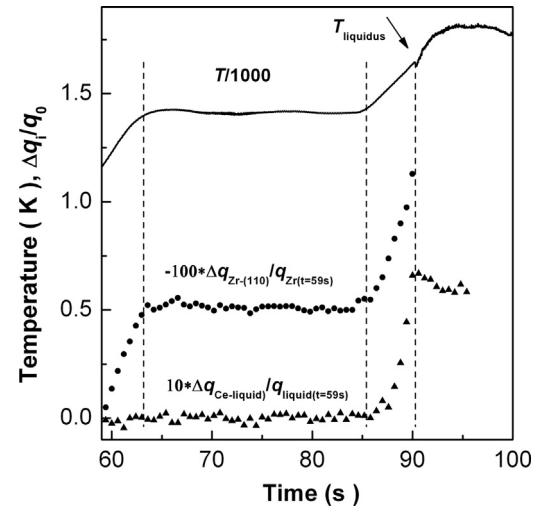

Fig. 2. Relative change of the positions of the (110)-reflection of *bcc*-Zr (circles) and the diffuse maximum of the Ce-rich liquid (triangles) in comparison with temperature profile (solid line). The shift of the (110)-reflection reflects the thermal expansion and corresponds to the temperature profile. The shift of the diffuse maximum also clearly responds to the temperature profile for $T > 1300$ K.

The liquidus temperature can be seen by the slope change of the pyrometer data upon heating. Also the crystallization during the cooling cycle is evident by the recalescence in Fig. 1. The crystallization upon cooling starts at a lower temperature than the liquidus temperature upon heating due to undercooling of the melt. For the data evaluation in the following only XRD patterns of heating cycles were used. The heating rate was usually about 50 K/s. This means that the XRD patterns each recorded for 0.6 s represent an average over a temperature span of 30 K. The accuracy of temperature measurement by the pyrometer is given by 1% of reading. A value of $\Delta T = \pm 20$ K estimated as error for the temperature of the *in situ* diffraction data. In order to check whether the diffraction data can be related to the corresponding equilibrium states, the extracted structural information of the second heating cycle in Fig. 1 were analysed in detail. Fig. 2 compares the temperature profile of $Ce_{40}Zr_{60}$ with the observed relative changes $\Delta q_i/q_i$ of two characteristic diffraction peak positions q_i, which are the position of the diffuse maximum of the liquid phase $q_{Ce(Zr)\text{-}liquid}$ and the position of the (110) reflection of the *bcc*-Zr(Ce) phase $q_{Zr\text{-}(110)}$. A clear correlation is visible between temperature profile and diffraction data. The shift of $q_{Zr\text{-}(110)}$ reflects the thermal expansion of the *bcc*-Zr phase with an increasing temperature. For the range of constant temperature $T = 1300$ K between $t = 65$ s and $t = 85$ s we observe also a constant peak position at $q_{Zr\text{-}(110)} = (24.340 \pm 0.003)\ \mathrm{nm}^{-1}$. For the much broader diffuse maximum of the liquid we obtain $q_{Ce(Zr)\text{-}liquid} = (21.14 \pm 0.05)\ \mathrm{nm}^{-1}$ (the scaling of the relative position changes in Fig. 2 is different by a factor of 10). The clear shift of $q_{Ce(Zr)\text{-}liquid}$ at the higher temperature $T > 1300$ K is mainly related to change in chemical composition as discussed later. Above the liquidus temperature the composition of the liquid stays constant and the peak positions exhibit only some scatter. The close correlation of the structural data with the temperature profile as shown in Fig. 2 justifies to use them for the analysis of the phase equilibria within the error limits in temperature and peak positions.

3. Experimental results

Fig. 3 shows representive microstructures of an arc-melted $Ce_{40}Zr_{60}$ alloy by the SEM images. A homogeneous phase

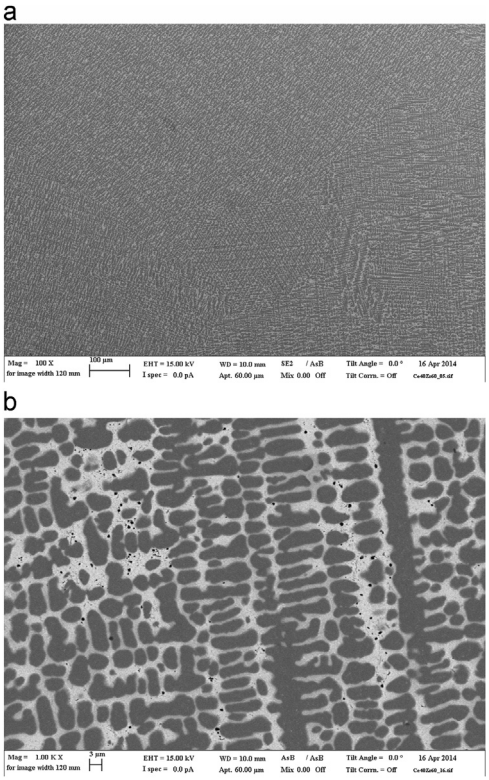

Fig. 3. SEM images of an arc-melted $Ce_{40}Zr_{60}$ alloy. Zr-dentrides (dark) and Ce-matrix phase (bright) are regular distributed over large sample area. (b) 10 times enlarged part of (a).

Table 1
Phases, crystal structure and transformation temperatures of the elements Ce and Zr.

Phases	Phases name [5,6]	Space group	Temperature (K)	Refs.
bcc-Ce	δ(Ce)	Im-3m	$T_{melt}=1071$	[12]
fcc-Ce	γ(Ce)	Fm-3m	$T_{fcc\leftrightarrow bcc}=1001$	[6]
dhcp-Ce	β(Ce)	$P6_3/mmc$	$T_{dhcp\leftrightarrow fcc}=283$	[12]
bcc-Zr	β(Zr)	Im-3m	$T_{melt}=2128$	[6]
hcp-Zr	α(Zr)	$P6_3/mmc$	$T_{hcp\leftrightarrow bcc}=1139$	[6]

distribution is observed over the whole sample area consisting of Zr-dendrites surrounded by a Ce-matrix phase as can be seen in Fig. 3(a). The enlarged image Fig. 3(b) clearly shows the dentritic microstructure. No indication of liquid phase separation could be seen in the optical and SEM images of the arc-melted Ce–Zr samples. The $Ce_{40}Zr_{60}$ sample consists of *fcc*-Ce and *hcp*-Zr as determined by the XRD measurement. The reported phases and crystal structures of the Ce–Zr system are summarized in Table 1. The solidification behaviour of the $Ce_{40}Zr_{60}$ alloy is obviously characterized by the primary crystallization of Zr from the Ce–Zr melt. Local EDX analysis of the dendrites and the matrix phase results in 2.6 ± 0.2 at% Ce dissolved in *hcp*-Zr and 0.3 ± 0.1 at% Zr dissolved in *fcc*-Ce. The lattice parameters of the Zr-phase $a=(0.3238\pm0.0001)$ nm and $c=(0.5158\pm0.0002)$ nm are increased compared to elemental Zr ($a=0.3232$ nm, $c=0.5148$ nm [13]) due to dissolved Ce. Accordently, the value $a=(0.5152\pm0.0002)$ nm for the *fcc*-Ce is slightly smaller than pure the elemental Ce ($a=0.5155$ nm [14]). It should be noted that a discrepancy exists for the stable phase of Ce at room temperature. The reported Ce–Zr phase diagram in [5–7] shows the double close-packed hexagonal Ce phase (*dhcp*-Ce) as room temperature phase transforming at $T_{dhcp\leftrightarrow fcc}=334$ K into the face-centered cubic Ce phase

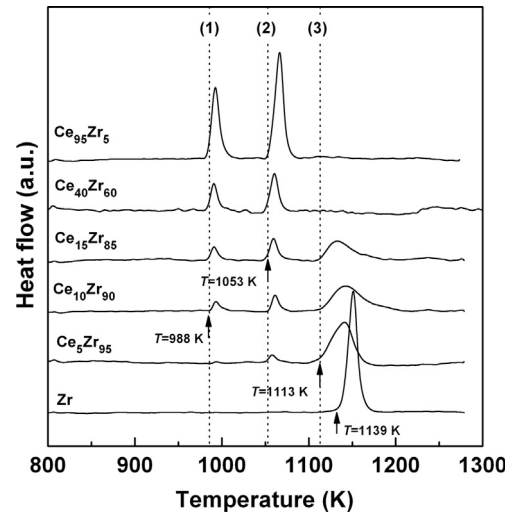

Fig. 4. DSC scans of different Ce–Zr alloys and that of pure Zr upon heating (heating rate 20 K/min). The on-set temperatures of the transformations are given.

(*fcc*-Ce). Detailed investigations of the phase transitions of elemental Ce found out that the *dhcp*-Ce phase slowly transforms to *fcc*-Ce phase at room temperature and a transition temperature $T_{dhcp\leftrightarrow fcc}=283$ K was determined [12]. This indicates that the *fcc*-Ce phase is the stable modification at standard temperature and pressure.

The thermal behaviour of different Ce–Zr alloys are shown in Fig. 4 by the DSC scans upon heating. All alloys undergo three phase reactions at identical temperatures as indicated by the exothermic events. The corresponding values are given in Fig. 4. The estimated error of the on-set temperatures is about ±5 K for the invariant reactions. The endothermic DSC events can be related to the solid phase transition (1) and the melting (2) of the Ce phase. Both transformation temperatures are about 10 K lower than for the elemental Ce. The phase reaction (3) is related to the solid state transformation of the Zr phase. The corresponding temperature in the Ce–Zr alloys is about 30 K less than the value for pure Zr as shown in Fig. 3. For the alloys with Zr-content ≤60 at% the DSC event (3) was not detected probably due to reaction of the liquid Ce phase with the corundum container. This is also the reason for essentially altered DSC curves upon cooling (not shown).

To overcome the problem of phase reaction and to have a direct proof of the crystalline structures, the phase equilibria were investigated by combining containerless levitation with *in situ* XRD measurements at elevated temperatures. Fig. 5 shows as an example continuously measured diagrams $I(q)$ of the $Ce_{40}Zr_{60}$ alloy during the heating and cooling cycles according to Fig. 1. The occurrence of solid phase transitions and melting of the Ce-phase can be clearly seen. The position of the diffuse maximum of the liquid phase changes with temperature as indicated by the dashed line in Fig. 5. The analysis of the diffraction peaks enables to determine directly the different coexistence ranges of phases as a function of temperature. Because no intermetallic phase is formed in the Ce–Zr system, we observe only the different allotropes of the elements and the liquid phase in the XRD patterns. Fig. 6 shows selected diagrams according to the phase sequences as a function of the temperature. The XRD patterns demonstrate that the different solid state transformations can be clearly distinguished and the transition temperatures from the pyrometer data are in a good agreement with the DSC events. Together with the DSC data we can conclude that below $T=988$ K, *fcc*-Ce(Zr) and *hcp*-Zr(Ce) coexist. The reaction (1) of the DSC curve at $T=988$ K corresponds to the transformation *fcc*-Ce + *hcp*-Zr ↔ *bcc*-Ce. Up to $T=1040$ K we

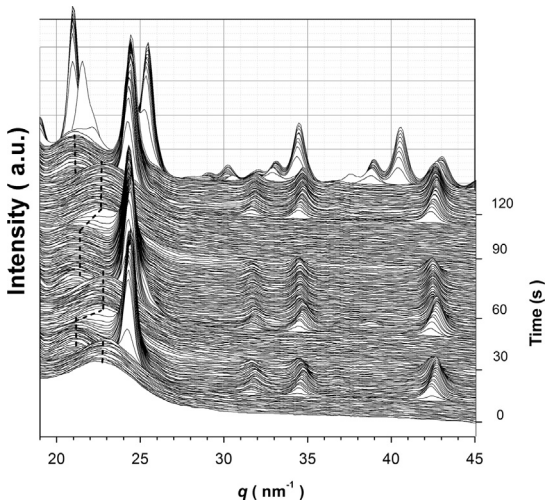

Fig. 5. *In situ* XRD patterns of $Ce_{40}Zr_{60}$ at elevated temperatures. The temperature varies according to the profile shown in Fig. 1. Phase transitions and melting of the sample is clearly visible. The position of the diffuse maximum of the liquid phase shifts towards larger q-values with temperature as indicated by the dotted line.

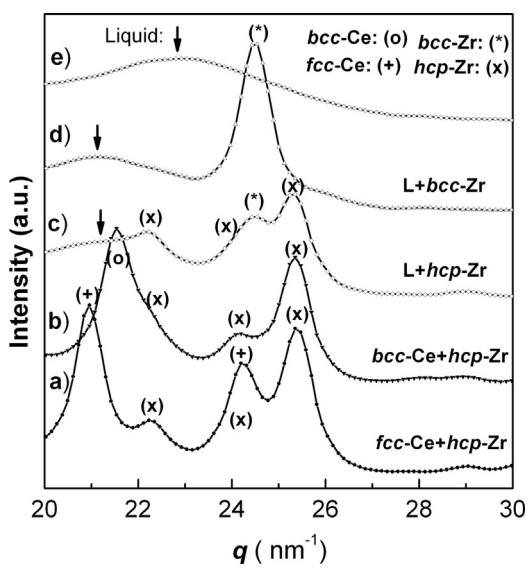

Fig. 6. *In situ* XRD patterns of $Ce_{40}Zr_{60}$ at different temperatures showing the phase sequences upon heating. (a) $T=950$ K, (b) T=1020 K, (c) $T=1100$ K, (d) $T=1200$ K, (e) $T=1700$ K. The reflections are marked according to the phase identification. The position of the diffuse maximum of the liquid phase is indicated by the arrows.

observe the reflections of *bcc*-Ce(Zr) and *hcp*-Zr(Ce). At $T=1053$ K, DSC event (2), the eutectic reaction *bcc*-Ce(Zr)+*hcp*-Zr(Ce)↔L takes place. The partly melting of the sample can be seen by the disappearance of the *bcc*-Ce(Zr) reflections and the occurrence of a diffuse diffraction peak in the XRD-patterns. Between 1053 K < T < 1100 K we observe the equilibrium between Ce-rich liquid and *hcp*-Zr(Ce) phase. The DSC event (3) at $T=1113$ K corresponds to the phase reaction L+*hcp*-Zr(Ce)↔*bcc*-Zr(Ce). Above $T>1113$ K the Ce-rich melt is in equilibrium with the *bcc*-Zr(Ce) phase. For the $Ce_{40}Zr_{60}$ alloy, the fraction of *bcc*-Zr(Ce) decreases with temperature up to $T=1640$ K, and above $T_{liquidus}=(1650\pm20)$ K only liquid phase is present (Fig. 6). In the semi-solid regime one can observe a systematic shift of the position of the diffuse maximum of the liquid phase indicated by the arrows in Fig. 6.

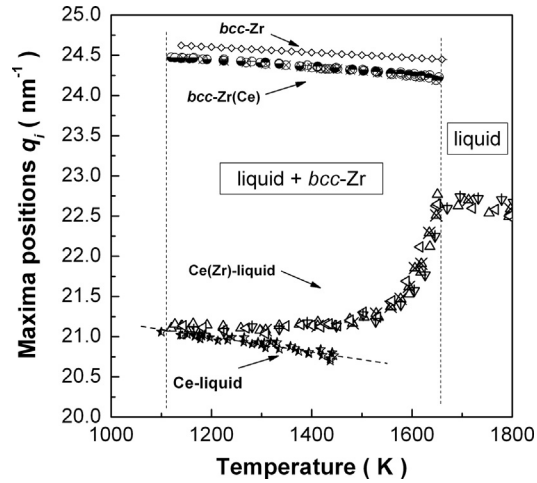

Fig. 7. Positions of diffraction maxima q_i of $Ce_{40}Zr_{60}$ as a function of emperature. Triangles: q_1(T) of Ce(Zr) liquid phase; circles: q_1(T)of *bcc*-Zr(Ce). Also show are the positions of the pure elements. Stars: q_1(T) of Ce-liquid; diamonds: q_1(T) of *bcc*-Zr.

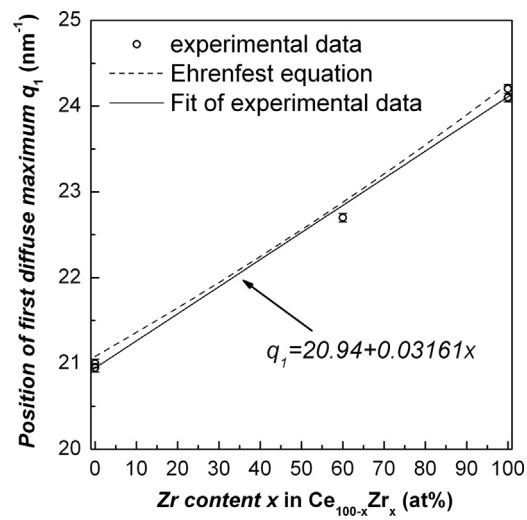

Fig. 8. Composition dependence of the position of the first diffuse maximum q_1 of $Ce_{100-x}Zr_x$ liquids (x=0, 60, 100). Analytical functions by a linear fit of experimental data (line) and by the Ehrenfest-equation (dashed line) are given.

The positions of the diffuse maxima of the liquid phase and of the (110)-reflection of the *bcc*-Zr phase were calculated by a fit of two Lorentzian curves. Fig. 7 shows the temperature dependence of both peak positions q_i together with data measured for the pure elements (Ce liquid and *bcc*-Zr). The presented data were obtained from independent measurements of four different heating cycles in order to prove the reproducibility. The shift of q_1 of the Ce(Zr) melt towards larger q-values reflects mainly the change in composition with temperature and contains the information about the liquidus line. For metallic liquids and glasses the position of the first diffuse maximum q_1 scales approximately with a concentration-averaged atomic diameter d of the alloy described by the Ehrenfest-equation $q_1=1.22\times2\pi/d$ with $d=\sum c_i\times d_i$, where d_i is the atomic diameter of element i and c_i is the concentration of element i [15]. This equation derived for gas molecules is not exactly valid for condensed matter, however, within an element substitution series the change in q_1 can be often well described by a linear dependence on the concentration c_i of the added element [16,17]. The experimentally observed shift of the position of the maxima to larger q-values with rising temperature then can be understood by an increased solubility of Zr ($d_{Zr}=0.316$ nm) in the Ce-rich liquid ($d_{Ce}=0.3636$ nm) which

reduces the average atomic diameter. Fig. 8 shows the experimentally determined q_i values for the melts of Ce, Zr, and $Ce_{40}Zr_{60}$ as well as the calculation by the Ehrenfest-equation. From these data we can conclude, that the concentration dependence of the maximum position q_1 *versus* Zr content x can be well approximated by a linear function as given in Fig. 8. The thermal expansion of the liquids has also an impact on the maxima positions as can be seen by the behaviour of $q_{Ce\text{-}liquid}$ of the pure Ce melt in Fig. 7. We observe a small shift towards lower q-values which can be approximated by $q_{Ce-liquid}(T) = (21.12 - 9.38 \times 10^{-4}T)\text{nm}^{-1}$. Assuming the same thermal coefficient for the Ce–Zr liquid, we can correct the effect of volume expansion on the maximum position in order to extract the compositional contribution. Using the composition dependence according to the fit function shown in Fig. 8, we can calculate the Zr-content x in at% of the Ce(Zr) liquid at a given temperature from the experimental data $q_1(T)$ in nm^{-1}:

$$x_{Zr}^{liquid}(T) = \frac{q_1(T) - 21.12 + 9.38 \times 10^{-4}\,T}{0.03161} \tag{1}$$

By applying Eq. (1) for the values of the liquid diffraction maxima in the two phase region L+*bcc*-Zr(Ce), the calculated compositions then represent the liquidus line of the Ce–Zr phase diagram, as shown in Fig. 9 by the data points. Concerning the error limit in the peak positions of the diffuse maxima $\Delta q = \pm 0.05\ \text{nm}^{-1}$, the error in the calculated composition of the Ce(Zr) liquid is about ± 2 at%. From the temperature dependence of the liquidus line $x_{Zr}(T)$, the eutectic composition can be determined as 1.5 ± 1 at% Zr.

Similar to the liquid phase one can also calculate the solubility of Ce in the *bcc*-Zr(Ce) phase by the analysis of the difference of the peak positions to those of the pure *bcc*-Zr. Using Vegard's rule for a solid solution between *bcc*-Ce and *bcc*-Zr, the concentration of Ce (in at%) of the *bcc*-Zr(Ce) phase is given by:

$$x_{Ce}(T) = \frac{q_{bcc-Zr(Ce)}(T) - q_{bcc-Zr}(T)}{q_{bcc-Ce} - q_{bcc-Zr}} 100 \tag{2}$$

The temperature dependence of the q-value of the pure *bcc*-Zr phase was calculated by an extrapolation of the literature data lattice parameter given in [18], also presented in Fig. 5. For the difference $\Delta q = q_{bcc-Ce} - q_{bcc-Zr}$ of the pure *bcc*-phases a constant value $\Delta q = 3.03\ \text{nm}^{-1}$ is used according to values $a_{bcc\text{-}Zr} = 0.3609$ nm at $T = 1135$ K [18] and $a_{bcc\text{-}Ce} = 0.4135$ nm measured at $T = 1000$ K. A maximum solubility of 8 ± 1 at% Ce in *bcc*-Zr(Ce) at

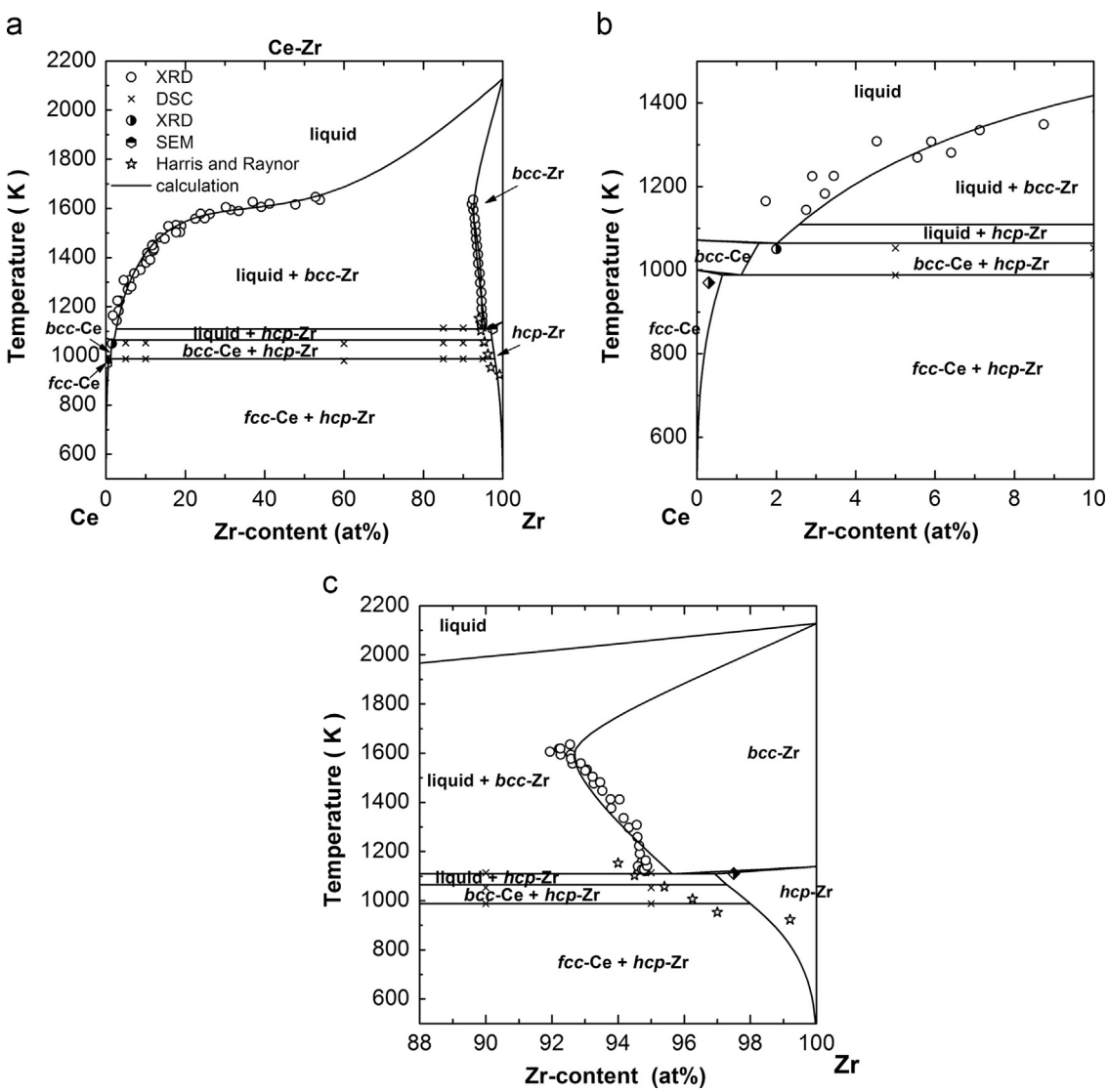

Fig. 9. (a) Ce–Zr phase diagram. Experimental data from *in situ* XRD, DSC and SEM are indicated by marks. Data of Ce-solubility in *hcp*-Zr from literature [8] are given by stars. Calculation using thermodynamic description of this work is represented by the solid lines. (b), and (c) show enlarged parts of the phase diagram (a).

$T=1500$ K is obtained from the diffraction data. The lattice parameter of the *bcc*-Ce(Zr) phase at $T=1050$ K, $a_{bcc\text{-}Ce(Zr)}=0.4123$ nm, is slightly smaller than that of pure *bcc*-Ce and the solubility of about 2 ± 1 at% Zr in *bcc*-Ce is calculated from the difference. Within the error limits identical values were obtained for the lattice parameters of the *fcc*-Ce phase in $Ce_{40}Zr_{60}$ and elemental Ce and ($a_{fcc}=0.5197$ nm at $T=1000$ K), which is in accordance with the low solubility of 0.3 at% Zr as determined by the EDX analysis. The experimental results, transformation temperatures determined by the DSC measurements, solubility by SEM analysis and XRD, phase compositions and liquidus line according to the *in situ* XRD, are summarized in the phase diagram shown in Fig. 9.

4. Thermodynamic calculations

The thermodynamic assessment of the binary Ce–Zr system was done by the CALPHAD method [4]. The Gibbs energy of each element was defined with respect to its stable magnetically disordered state at 298.15 K and 10^5 Pa (stable element reference). In the SGTE-Pure database 4.4 [19], the Gibbs energy was represented as a power series in terms of the temperature in the form:

$$^0G(T)=AT\ln(T)+\sum_{n=-1}C_nT^n, \tag{3}$$

where A, and C_n were coefficients and n represents a set of integers. The reference states for pure cerium and zirconium are fcc-Ce and hcp-Zr, respectively. It should be noted that there are no data for dhcp-Zr available in the SGTE pure database. The difference between hcp-Zr and dhcp–Zr was chosen as 1000 J/mol, which is approximately the same value as in the Ce system.

The liquid and the disordered solid solutions of the *bcc*, *fcc*, *hcp* and *dhcp* phases were described using a substitutional solution model, the Gibbs free energy being defined as:

$$G^{\phi}=\sum_{i=1}^{2}x_i{}^0G_i+G^{id}+G^{ex};\quad G^{id}=RT\sum_{i=1}^{2}x_i\ln(x_i);$$
$$G^{ex}=\sum_{i=1}^{2}\sum_{j>i}^{2}x_ix_j\sum_{v=0}^{n}{}^vL_{i,j}(x_i-x_j)^v, \tag{4}$$

where ^0G_i is the Gibbs energy of the pure element i, i.e., Ce or Zr in the corresponding phase φ, G^{id} is the ideal mixing contribution, G^{ex} is the excess energy, x_i and x_j are the mole fractions of the constituents i, and j, respectively. $^{\nu}L_{i,j}$ is the binary interaction parameter of components i and j of an order ν. R is the gas constant, and T is the absolute temperature.

Table 2
Summary of thermodynamic parameters describing the Ce–Zr system referred to the stable element reference H^{SER} ($T=298.15$ K, $P=100$ kPa). Values are given in SI units (J, mol, and K).

Parameter	Refs.
$G^{hcp}_{Ce}, G^{dhcp}_{Ce}, G^{fcc}_{Ce}, G^{bcc}_{Ce}, G^{liquid}_{Ce}, G^{hcp}_{Zr}, G^{bcc}_{Zr}, G^{fcc}_{Zr}, G^{liquid}_{Zr}$	[19]
Liquid	
$^0L^{liq}_{Ce,Zr}=+21952.51$	This work
$^1L^{liq}_{Ce,Zr}=+5384.14$	This work
hcp	
$^0L^{hcp}_{Ce,Zr}=+33078.5$	This work
bcc	
$^0L^{bcc}_{Ce,Zr}=+26480.74+7T$	This work
$^1L^{bcc}_{Ce,Zr}=+3766.75$	This work
fcc	
$^0L^{fcc}_{Ce,Zr}=+35039.6$	This work
dhcp	
$G^{dhcp}_{Zr}=G^{hcp}_{Zr}+1000$	This work
$^0L^{dhcp}_{Ce,Zr}=+70,000$	

Table 3
Invariant reactions of the Ce–Zr system.

	T (K)		Calc. compositions of phases (at% Zr)		
	calc	exp	1	2	3
bcc-Zr ↔ L + hcp-Zr	1109	1113	95.64	2.57	96.92
L ↔ bcc-Ce + hcp-Zr	1065	1053	2.03	1.56	97.29
bcc-Ce ↔ fcc-Ce + hcp-Zr	988	988	1.12	0.64	97.99

The thermodynamic model parameters for the binary system Ce–Zr were evaluated on the basis of the experimental information of this work using the least-squares optimization module Parrot of the software Thermo-Calc [20]. First mixing parameters of *bcc*, *hcp* and liquid phase were assessed to fit the liquidus and solidus lines and the two invariant reactions *bcc-Zr* ↔ *L* + *hcp*-Zr and *L* ↔ *bcc*-Ce + *hcp*-Zr from the present work. Then mixing parameter of *fcc* was evaluated to fit the third invariant reactions *bcc*-Ce ↔ *fcc*-Ce + *hcp*-Zr. The final set thermodynamic parameters was obtained by simultaneous optimisations of all mixing parameters. The results are presented in Table 2. The computed Ce–Zr phase diagram using the present set of thermodynamic parameters is shown in Fig. 9 along with the experimental results. The calculated temperatures and phase compositions of the invariant equilibria are given in Table 3.

5. Discussion

The present results of modelling agree well with the experimental data obtained in the present work. The calculated maximal solubility of 7.2 at% Ce in *bcc*-Zr at 1634 K is in good agreement with measured solubility of 8 at% Ce. Also, the calculated maximal solubility of Ce in *hcp*-Zr 3.1 at% is in a good accordance with the value of 2.5 at% determined by the EDX/SEM analysis of the arc-melted $Ce_{40}Zr_{60}$ sample. The calculated temperature (1113 K) and Ce solubility in *bcc*-Zr (4.4 at%) for the catatectic reaction *bcc*-Zr ↔ *L* + *hcp*-Zr agrees well with the present experimental results of 1109 K and 5 at%, respectively. Calculated maximal solubility of Zr in the *bcc*-Ce phase equal to 1.6 at% which is close to the value of 2.0 at% measured in the present work. Maximal solubility of Zr in the *fcc*-Ce phase is 0.56 at% according to calculations agrees well with 0.35 at% from EDX/SEM analysis of the arc-melted $Ce_{40}Zr_{60}$ sample. Furthermore, the phase equilibrium between *dhcp*-Ce and *hcp*-Zr for $T<283$ K (not shown in Fig. 9) is reproduced by the derived thermodynamic description.

The phase diagram of this work as shown in Fig. 9 is in contradiction to the earlier reports [5–7]. In our opinion the lack of reliable thermodynamic data had led to the wrong presentation as monotectic system in the past. By the *in situ* XRD measurements of this work clear evidence is found for the eutectic character of the Ce–Zr system, which is similar to the Gd–Zr [21] and Nd–Zr [22] phase diagrams. Also the data of solubility of Ce in *hcp*-Zr as reported in [8] are to some extent inconsistent. There is no calibration curve given, the lattice parameter as function of composition exhibits a quite strange slope change and no information is given whether or not the samples are single phase after annealing. According to our findings, the reported highest annealing temperature $T=1123$ K is in the *bcc*-Zr state. Because the addition of Ce lowers the *hcp*-Zr ↔ *bcc*-Zr transformation temperature in comparison with pure Zr, the Ce solubility in the *hcp*-Zr should be less than in *bcc*-Zr measured as 5 at% in the present study, which is in accordance with the SEM/EDX analysis of the *hcp*-Zr dendrites of the $Ce_{40}Zr_{60}$ arc-melted alloy.

6. Conclusions

Experimental and thermodynamic assessment of the binary Ce–Zr system has been carried out. Combining conical nozzle levitation technique with *in situ* high-energy synchrotron X-ray diffraction enabled to observe directly the phase equilibria at elevated temperatures. Especially for alloy systems with high liquidus temperature, and/or reactive melts the used method has substantial advantages in determination of phase sequences and range of coexistence with the liquid phase. For binary metallic systems the liquidus line can be determined from the diffraction data of the liquid phase. The experimental observations led to a complete new Ce–Zr phase diagram, which is found to be quite similar to those of recently assessed Gd–Zr and Nd–Zr systems [21,22]. Based on the experimental results, a consistent thermodynamic description was developed.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2014.05.002.

References

[1] M.J. Kramer, R.W. McCallum, I.A. Anderson, S. Constantinides, JOM 64 (7) (2012) 752–763.
[2] M. Ozawa, J. Alloys Compd 275–277 (1998) 886.
[3] R.C.R. Netob, M. Schmala, Appl. Catal. A: Gen. 450 (2013) 131–142.
[4] H.L. Lukas, S.G. Fries, B. Sundman, Computational Thermodynamics – The Calphad Method, Cambrige University Press, 2007.
[5] W.G. Moffatt: "Binary Phase Diagrams Handbook", Schenectady, N.Y. General Electric Comp. (1987).
[6] T.B. Massalski (Ed.), Binary Alloy Phase Diagrams, 2nd ed., American ASM International, Materials Park, OH, 1990.
[7] A. Palenzona, S. Cirafici, J. Phase Equilib. 12 (1991) 49–52.
[8] I.R. Harris, G.V. Raynor, J. Less-Common Met. 6 (1964) 70–80.
[9] A. Mizuno, J. Tamura, S. Kohara, M. Watanabe, ISIJ Int. 52 (2012) 770–773.
[10] P.F. Paradis, T. Ishikawa, G.W. Lee, D. Holland-Moritz, J. Brillo, W.K. Rhim, Junpei T. Okada, Mater. Sci. Eng. R 76 (2014) 1–53.
[11] A.P. Hammersley, S.O. Svensson, M. Hanfland, A.N. Fitch, D. Häusermann, High Press. Res. 14 (1996) 235.
[12] K.A. Gschneidner, V.K. Pecharsky, J. Phase, Equilibria 20 (6) (1999) 612–614.
[13] J. Goldak, L.T. Lloyd, C.S. Barrett, Phys. Rev. 144 (2) (1966) 478–484.
[14] V.S. Pavlow, V.A. Finkel, Phys. Met. Metallogr. 24 (6) (1967) 137–138.
[15] P. Ehrenfest, in: Proceedings of Koninklijke Akademie van Wetenschappen Amsterdam 17, 1915, pp. 1184–1190.
[16] K.H.J. Buschow, J. Phys. F: Met. Phys. 14 (1984) 593–607.
[17] N. Mattern, A. Schops, U. Kühn, J. Acker, O. Khvostikova, J. Eckert, J. Non-Cryst. Solids 354 (2008) 1054–1060.
[18] R.B. Russell, J. Met. 6 (1954) 1045.
[19] SGTE Unary Database version 4.4 ⟨www.sgte.org⟩.
[20] Thermocalc Software ⟨www.thermocalc.com⟩.
[21] N. Mattern, J.H. Han, M. Zinkevich, O. Shuleshova, I. Kaban, D. Holland-Moritz, J. Gegner, F. Yang, J. Bednarcik, W. Loser, J. Eckert, Comput. Coupling Phase Diagr. Thermochem. 39 (2012) 27–32.
[22] N. Mattern, Y. Yokoyama, A. Mizuno, J.H. Han, O. Fabrichnaya, T. Harada, S. Kohara, J. Eckert, Comput. Coupling Phase Diagr. Thermochem. 46 (2014) 103–107.